

Bigya Tuladhar

Senior Full-Stack & Cloud Engineer | 8+ Years | AWS Serverless & Infrastructure as Code | RAG & Agentic AI Workflows | Real-Time Systems
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SUMMARY

Senior full-stack and cloud engineer with 8+ years of experience, currently the sole owner of the AI platform at Nova Dynamic Media, a live-events company. I care about reliability under real pressure over features that only work in a demo, and I've led development teams of 3 to 8 people while staying hands-on with architecture and code rather than stepping back into pure management.

TECHNICAL SKILLS

Languages & Frameworks: TypeScript, JavaScript, Python, Node.js, React, Next.js, Vue.js, NestJS, Express, FastAPI, Tailwind, Flutter
AWS & Cloud: AWS Lambda, API Gateway, AppSync, Step Functions, Kinesis, Glue, Athena, S3, DynamoDB, EC2, CloudFront, IAM, SQS, SNS, Amazon IVS, MediaConvert, KMS, Amazon Bedrock, ECR, Amplify, CloudWatch, VPC, GuardDuty, WAF, AWS Backup, AWS CDK, Serverless Framework, CloudFormation, Terraform
AI & Machine Learning: RAG (Retrieval-Augmented Generation), Knowledge Bases, Vector Databases, Semantic Search, Embeddings, Agentic Workflows, LangGraph, Bedrock AgentCore, Prompt Engineering, LLM Integration (Claude, Amazon Nova, DeepSeek), Grounding and Guardrails, Cross-Region Inference Failover
Data, DevOps & API: PostgreSQL, MongoDB, MySQL, DynamoDB, Redis, Valkey, AWS CodeBuild, Jenkins, CI/CD, Docker, GraphQL, REST API, Cognito, Mailgun
Practices: Serverless Architecture, Event-Driven Architecture, Infrastructure as Code (IaC), Microservices, System Design, Technical Leadership, Load Testing (K6, JMeter)

EXPERIENCE

Senior Full-Stack & Cloud Engineer — Nova Dynamic Media, Hong Kong (Remote) · Feb 2022 - Present

Engaged via Swivt Technologies through Dec 2024; joined Nova full-time Jan 2024.

- Built a production RAG platform on AWS Bedrock (S3 Vectors, hierarchical chunking, semantic search) that powers a live moderator copilot and post-event AI assistants, answering audience questions with source-backed accuracy across live events.
- Cut throttling failures 60-70% under burst load by engineering cross-region Bedrock inference failover with exponential backoff, keeping the live copilot responsive at peak.
- Own the platform's agentic AI layer end to end: a natural-language event-metrics agent (query generation, validation, and reasoning steps), automated QnA grouping, and AI report generation, now running on Amazon Bedrock AgentCore with infrastructure provisioned through Terraform.
- Built a real-time telemetry pipeline streaming live event activity through Kinesis Firehose into S3, with Valkey for real-time stats and Glue + Athena for post-event reporting and AI insights.
- Built real-time interaction services on AWS AppSync and DynamoDB, including a live QnA mode where audience votes update question rankings instantly, broadcasting real-time changes to 1,000+ concurrent viewers alongside polls, surveys, and synced slideshows, migrating the platform off AWS IoT Core in the process.
- Built the platform's mass email system: SQS queues import users into the platform, and a Step Functions workflow adds them to Mailgun mailing lists, our primary email provider, with AWS SES as backup.
- Delivered ultra-low-latency streaming via Amazon IVS over WebRTC with live transcription and translation, transcoded 1,000+ videos through MediaConvert, and built Mission Control, a React/TypeScript dashboard giving producers live control over streaming, QnA, and audience engagement during events.
- Cut AWS spend ~40% by moving CI/CD from Jenkins to AWS CodeBuild with multi-account, cross-region deployments, and shifted the platform toward serverless with CDK-based Infrastructure as Code.
- Built a QR-based event check-in app in Flutter with attendance reporting, and delivered custom event sites for 30+ clients with automated domain setup.

Co-Founder — Tissha Cosmetics (tissha.com) · Dec 2025 - Present

- Co-founded a cosmetics e-commerce company from the ground up, leading a team of 3 engineers and setting up VPS servers, infrastructure, and architecture while driving the platform from concept to production.

Senior Full-Stack Developer — Bitsky (Remote) · 2024

- Integrated third-party game provider APIs (session management, webhook callbacks, real-time balance updates) and built a secure Stripe wallet system in NestJS for deposits, payouts, and reconciliation.

Senior Full-Stack Developer / Web Team Lead — Swivt Technologies, Lalitpur, Nepal · Jan 2022 - Dec 2023

Agency role; primary engagement was leading the Nova platform (above).

- Led 5-8 developers across 5+ projects, coordinating sprints and delivery with the project manager; also built a custom CMS serving multiple client sites.
- Built Crystal Academy, an online e-learning platform with courses, tests, and certificates, plus e-commerce platforms including a Taekwondo studio store and Aitken Vanson (shoe retailer) with Stripe-integrated booking.
- Built e-commerce platforms with Stripe payment integration and a Google Calendar booking system.
- Introduced CI/CD pipelines to an agency with no established deployment process, replacing manual releases with automated builds and reducing the regression rate across client projects.
- Managed hosting for all client projects on GoDaddy, optimizing frontend delivery through Cloudflare caching, edge configuration, and DNS/CDN routing.

Full-Stack Developer — Upaya, Kathmandu, Nepal · Jun 2018 - Jan 2022

- Built an accounting and CRM system with role-based access and audit trails, a frontend search over 1M+ records running in 0.3-0.7 seconds, and automated reporting that saved 15+ hours a week; also shipped 5 hybrid mobile apps with Cordova.
- Built an end-to-end logistics platform (order intake, routing, proof of delivery) with live GPS tracking (+40% on-time delivery), a dynamic pricing algorithm based on route and vehicle type, and a customer service portal to speed up order processing.

SELECTED PROJECTS

Nova Dynamic Media (novaweb.live). Tissha Cosmetics (tissha.com). Aitken Vanson (aitkenvanson.com). Upaya (upaya.com.np, ubs.com.np). Crystal Academy (crystalacademy.org). Bitsky (bitsky.bet).

EDUCATION

Bachelor's Degree in Information Management, Tribhuvan University, 2014 - 2018